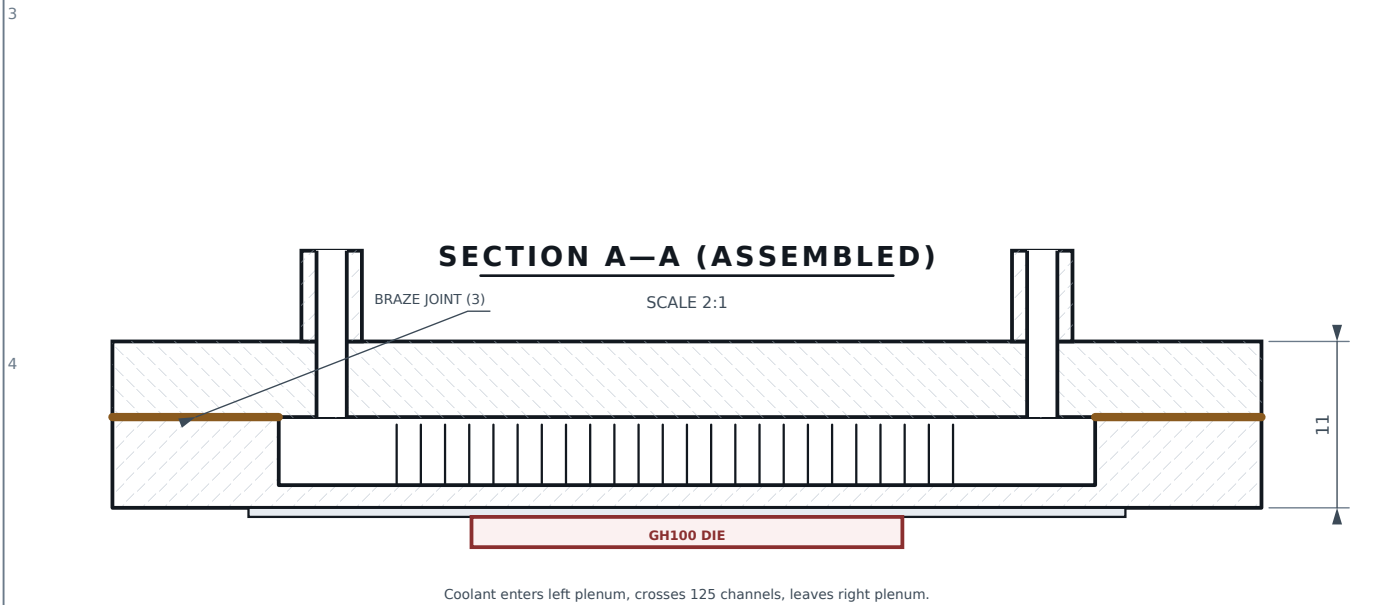
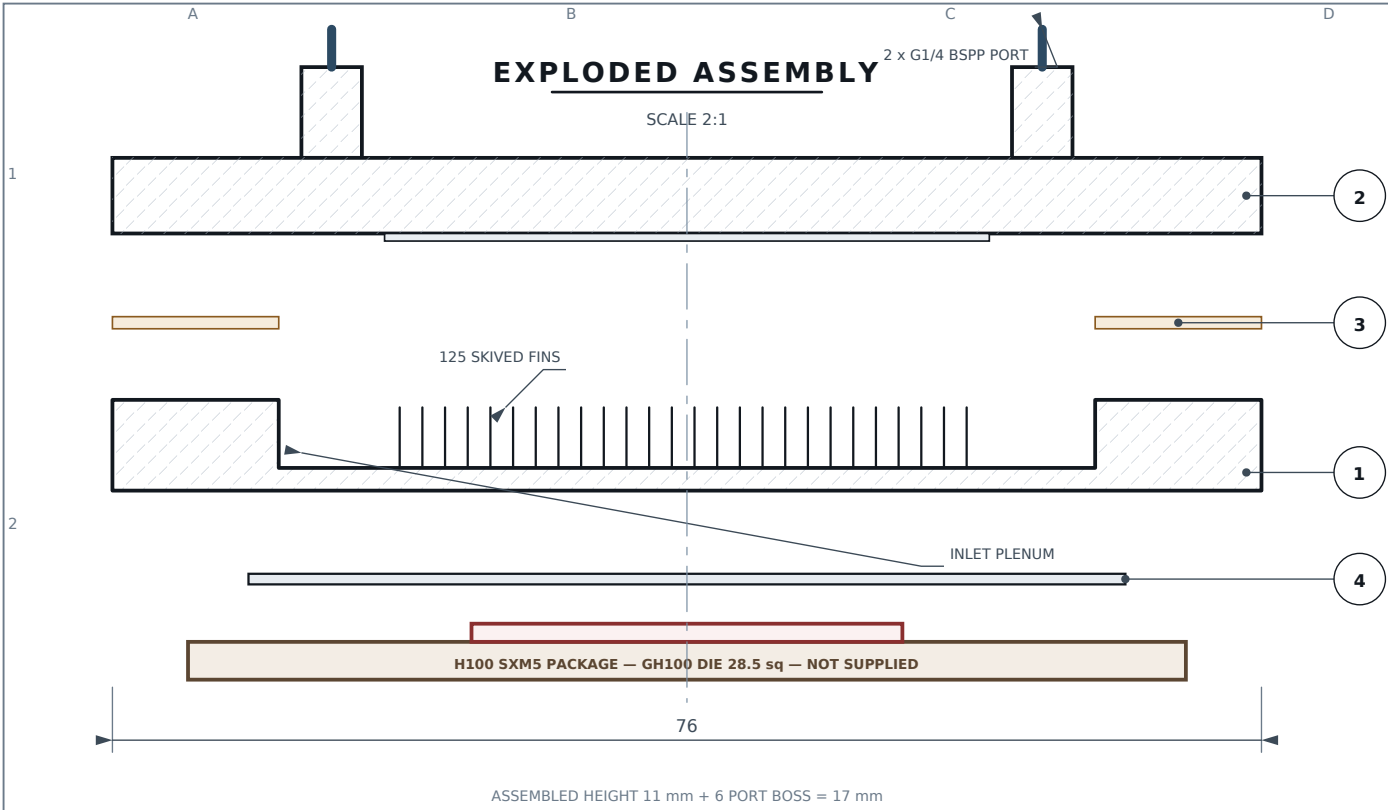


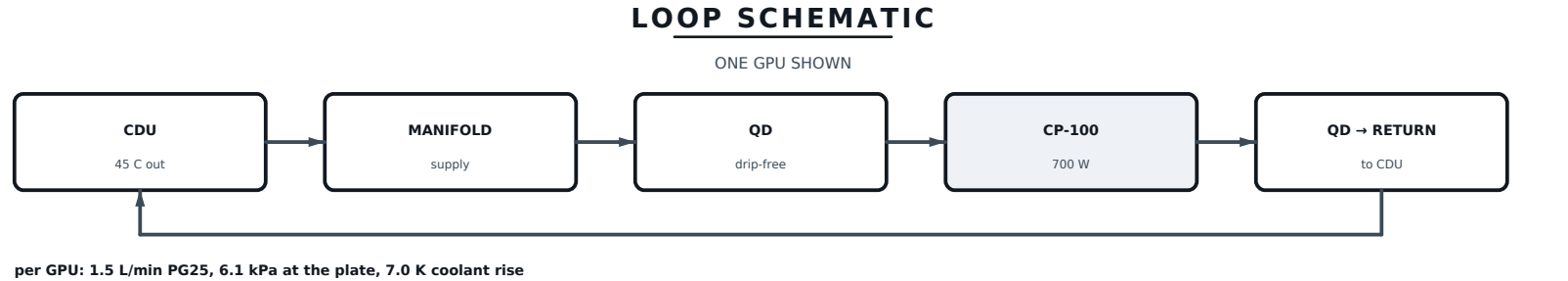


- GENERAL NOTES
- Interpret per ISO 128. Tolerances ISO 2768-mK unless stated. Dimensions in mm.
 - EVERY WETTED PART IS COPPER OR STAINLESS. No aluminium anywhere in the loop — a Cu/Al couple in glycol corrodes the aluminium out within months.
 - Cover is vacuum brazed to the body: no elastomer inside the loop, no serviceable joint. Replace, do not open.
 - SXM5 interface dimensions and mounting preload on sheet 5 are ASSUMED. Confirm against NVIDIA's mechanical specification before manufacture.

BILL OF MATERIALS — ASSEMBLY CP-100					
ITEM	QTY	PART No.	DESCRIPTION	MATERIAL	NOTE
1	1	CP-101	COLD PLATE BODY, SKIVED FIN	Cu C10200 / C1020	sheet 2
2	1	CP-102	COVER / MANIFOLD, 2 PORTS	Cu C10200 / C1020	sheet 3
3	1	CP-103	BRAZE PREFORM, PERIMETER RING	BCuP-5 (Ag 15%)	0.10 foil
4	1	CP-104	THERMAL INTERFACE, BARE DIE	PTM, k >= 13 W/mK	60 um BLT
5	4	CP-201	SPRING SCREW + BELLEVILLE STACK	A2-70 / 17-7PH	sheet 5
6	2	CP-202	QUICK DISCONNECT, DRIP-FREE	Ni-plated brass	loop side

RESISTANCE BUDGET — PG25, 1.5 L/min			
PATH	K/W	dT @ 700 W	SHARE
coolant (convection + sensible)	0.0141	9.9 K	48 %
TIM, bare die 60 um k=13	0.0057	4.0 K	19 %
spreading, die to fin field	0.0050	3.5 K	17 %
base 1.5 mm Cu	0.0047	3.3 K	16 %
TOTAL, die to coolant inlet	0.0295	20.7 K	100 %

DIE TEMPERATURE — SAME PLATE, 1.5 L/min				
COOLANT IN	H100 700 W	B200 1000 W	GB200 1200 W	
32 C	53 C	62 C	67 C	
40 C	61 C	70 C	75 C	
45 C	66 C	75 C	80 C	
50 C	71 C	80 C	85 C	



INDIA-VC THERMAL HARDWARE					ISO 128 / ISO 2768-mK	
TITLE D2C COLD PLATE — GENERAL ASSEMBLY H100 SXM5 / 700 W / Cu C10200, VACUUM BRAZED / PG25 1.5 L/min			SCALE 2:1		UNITS mm	
			THIRD ANGLE PROJECTION			
DRAWING No. CP-100			REV A		SHEET 1 / 6	
DRAWN	CHECKED	MATERIAL	FINISH		DATE	
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES		2026-09-10	

SCALE 2:1



SCALE 5:1



SCALE 1.5:1

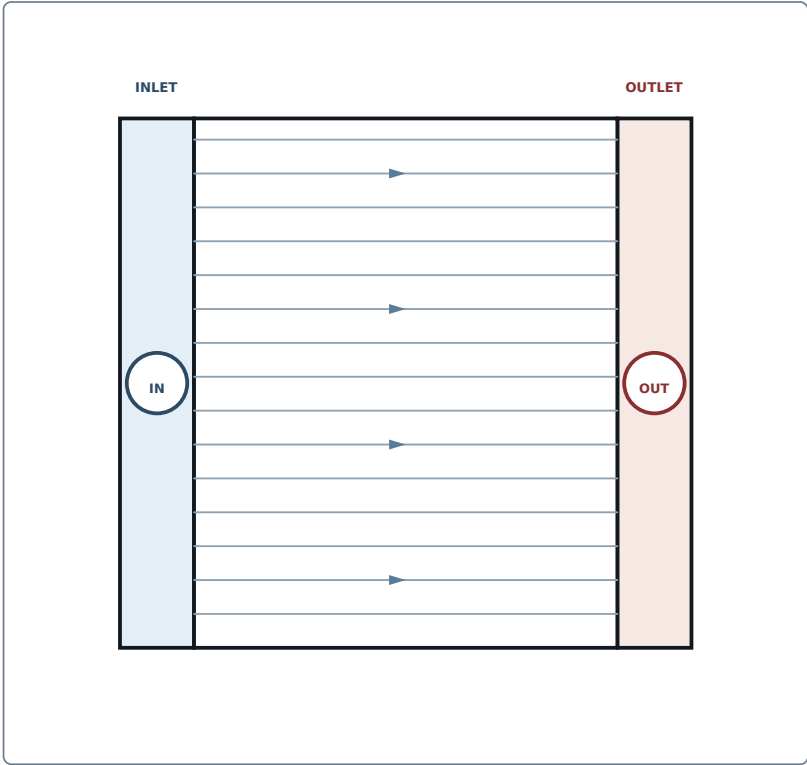


Ports open straight into the plenums; no internal turn.

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FLOW PATH

SCALE 1.4:1



125 PARALLEL CHANNELS — 0.25 W x 4.0 H x 40 LONG

Single pass, straight. No U-turn, no serpentine:

DESIGN POINT — 1.5 L/min, 700 W		
PARAMETER	WATER	PG25
velocity in channel	0.20 m/s	0.20 m/s
Reynolds number	170	71
heat transfer coefficient	7641 W/m2K	5300 W/m2K
fin efficiency	0.47	0.55
coolant temperature rise	6.8 K	7.0 K
pressure drop, plate only	2.5 kPa	6.1 kPa

ROBUSTNESS — CONSTANT FLOW					
CHANNEL BLOCKAGE	OPEN	VELOCITY	R total	DIE @45 C	dP
all clear	125	0.20 m/s	0.0295	65.7 C	6.1 kPa
5 % blocked	118	0.21 m/s	0.0299	66.0 C	6.5 kPa
10 % blocked	112	0.22 m/s	0.0303	66.2 C	6.9 kPa
20 % blocked	100	0.25 m/s	0.0313	66.9 C	7.7 kPa

READING THESE NUMBERS

- PG25 is the design fluid. Water is shown only to bound the gain from inhibited water where freeze protection is not required — about 5 % on total resistance.
- A SINGLE blocked channel is not a thermal event. The 1.5 mm copper base carries its ~10 W sideways 0.4 mm to the neighbours for roughly 0.1 K.
- Progressive fouling across many channels is the real risk, and it shows up first as rising pressure drop, not as a rising die temperature.
- So instrument the loop for DIFFERENTIAL PRESSURE, not only temperature, and filter to 50 um at the CDU. Service trigger: dP 20 % above the commissioning baseline at the same flow.

FLOW SENSITIVITY — PG25, 700 W				
FLOW	R total	DIE @45 C	RISE	dP
0.75 L/min	0.0373	71.1 C	14.0 K	3.1 kPa
1.00 L/min	0.0333	68.3 C	10.5 K	4.1 kPa
1.50 L/min	0.0295	65.7 C	7.0 K	6.1 kPa
2.00 L/min	0.0278	64.4 C	5.2 K	8.2 kPa
3.00 L/min	0.0261	63.3 C	3.5 K	12.3 kPa

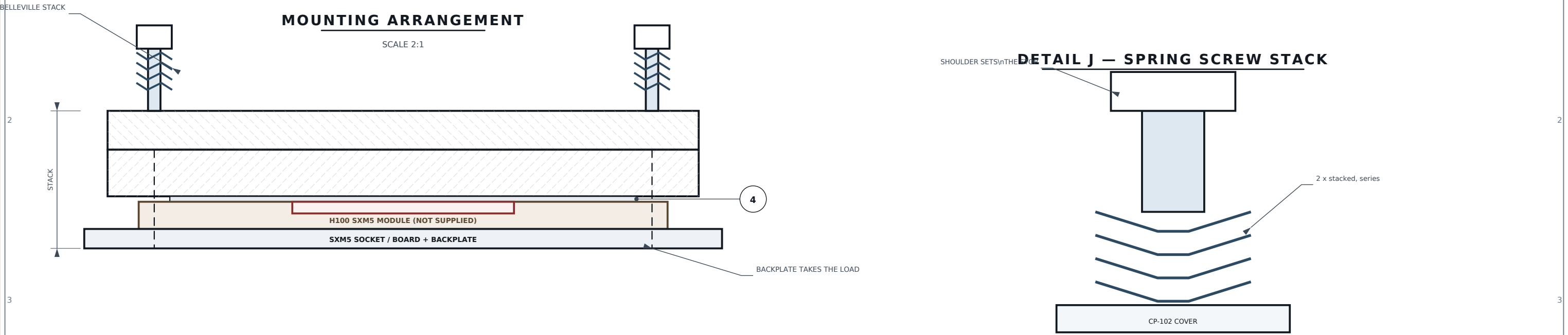
Below 1.0 L/min the plate loses ground quickly. Above 2.0 L/min it buys almost nothing — spend the pump head elsewhere in the loop.

INDIA-VC THERMAL HARDWARE					ISO 128 / ISO 2768-mK
TITLE THERMAL-HYDRAULIC SPECIFICATION H100 SXM5 700 W / PG25 / DESIGN POINT AND SENSITIVITIES			SCALE NONE	UNITS mm	THIRD ANGLE PROJECTION
DRAWING No. CP-103			REV A	SHEET 4 / 6	
DRAWN AI-ASSISTED DRAFT	CHECKED — PENDING —	MATERIAL SEE BOM	FINISH SEE NOTES	DATE 2026-09-10	

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ASSUMED INTERFACE — CONFIRM EVERY DIMENSION AND THE PRELOAD AGAINST NVIDIA'S SXM5 MECHANICAL SPECIFICATION BEFORE MANUFACTURE



MOUNTING PARAMETERS		
PARAMETER	VALUE	SOURCE
Total preload on the package	TO BE TAKEN FROM NVIDIA SPEC	NOT SET HERE
4 Screws	4 x M3 x 0.5, shouldered	this drawing
Spring element	2 x stacked, series	this drawing
Tightening	cross pattern, 3 passes to shoulder	this drawing
Cold plate flatness, die face	0.015 max	CP-101
Parallelism, plate to package	0.05 max	this drawing
TIM	PTM, k >= 13 W/mK, 60 um BLT	CP-104

The shoulder, not the torque wrench, sets the preload.

WHY THIS SHEET IS NOT FINISHED

Torque-only assembly on a bare die cracks silicon.

- An SXM5 module is a bare die on a socketed carrier. The socket, the load frame and the allowable preload are defined by NVIDIA, not by the cold plate designer.
- Too much preload cracks the die or the substrate. Too little starves the TIM bond line and the 4.0 K TIM budget on sheet 1 becomes 15 K or worse.
- Obtain the SXM5 mechanical and thermal design guide, then set the preload, the screw shoulder height and the Belleville stack to match it. Everything else on these six sheets is independent of that number.
- Do not substitute a torque specification for a shoulder stop. Torque on an M3 screw scatters by +/- 30 % with friction alone.

MOUNTING RISK REGISTER		
RISK	MECHANISM	CONTROL
Die crack	preload above the package rating, or uneven load	shoulder stop + cross tightening
High TIM resistance	insufficient preload, or air in the bond line	preload to spec, controlled dispense
6 TIM pump-out	thermal cycling with a soft grease	use a phase-change material
Plate rocking	flatness or parallelism out of tolerance	0.015 / 0.05 per this sheet

INDIA-VC THERMAL HARDWARE				ISO 128 / ISO 2768-mK	
TITLE		SCALE		UNITS	
SXM5 MOUNTING INTERFACE		2:1		mm	
LOAD FRAME, PRELOAD AND TIM / DIMENSIONS ASSUMED — SEE BANNER		THIRD ANGLE PROJECTION			
DRAWING No.		CP-200		REV	A
				SHEET	5 / 6
DRAWN	CHECKED	MATERIAL	FINISH	DATE	
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-10	

IN ORDER

through in months and the debris then blocks the 0.25 mm channels.

- Fill slowly from the lowest point. Vent at the highest point until clear.

- Service trigger: dP 20 % above the commissioning baseline at the same flow.

NONE

UNITSmmREV

A

SHEET6 / 6DRAWN

CHECKED

MATERIAL

FINISH

DATE _____AI-ASSISTED DRAFT

— PENDING —

SEE BOM

SEE NOTES2026-09-10